Design Brief – Review 2

Working draft at c32f2a4c0a54. Official code, owners and scope approval pending. Source means code evidence only; acceptance not run. See SRS/SDD.

0. Header

Project code	TBD (PKV working abbreviation)
Project name	PickleVerse – Multiplayer Pickleball Skill-Training and Match Game with AI Coaching
Team	Names and IDs deferred
Supervisor	Lại Đức Hùng (proposal)
Repository link (tag review-2)	github.com/Tad-Min/PickleVerse; HEAD c32f2a4; tag absent
Version	Draft 0.2 — not approved
Date	23 October 2026

1. Scope

1.1 Registration Changes

Proposed changes only; BM03 records the decisions requiring approval.

Section	Current	Proposed	Reason
Stack	Proposed Java/Python/UI Toolkit	Observed .NET/uGUI; approval pending	Iteration 1
Network	Dedicated proposed	Shared observed; no silent reduction	Trust boundary

1.2 Out of Scope

Item	Reason
Tournaments / spectators	Explicit proposal exclusion
Video swing analysis / ranked ladders	Explicit proposal exclusion
Real-money payment	Post-capstone extension

2. System at a Glance

PickleVerse helps players learn pickleball through guided controls, progressive drills and bot or online matches. Players review skill errors, progress, rewards and replay. Administrators manage users, content and service oversight. Current source links Unity to an ASP.NET Core backend with PostgreSQL, Redis, Photon Fusion Shared, SMTP and S3. Dedicated simulation, offline synchronization and evaluated ML remain gaps against the proposal. Architecture: SDD ARCH01 and ARCH02.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Identity	Local / API	Backend UUID prevents split identity.	M1	F01
D2	Awards	Client / ledger	Idempotent SQL commands prevent duplicate awards.	M4	F10
D3	Authority	Shared / dedicated	Shared exists; dedicated gap needs decision.	M3	F09
D4	Curriculum	Static / config	Versioned catalog; publish path incomplete.	M1	F03
D5	Data	Normalize / hybrid	Document actual SQL and JSONB boundaries.	M4	ERD00

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Peer trust and scope gaps	Trust/scope	Resolve SR01–03; stage adversarial tests.	M3
R2	Admin integration gaps	Blocked flows	Contract/status/publish tests SR04–07.	M4
R3	Unverified acceptance	Defense failure	Tag/CI/device evidence SR08–09.	M2

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
SR01	Peer authority differs	F09	Resolve; BM02	M3	TBD
SR02	Offline sync absent	F14	Resolve; BM02	M1	TBD
SR03	ML evaluation absent	F05	Resolve; BM02	M1	TBD
SR04	Web login DTO differs	F15	Resolve; BM02	M4	TBD
SR05	SQL status mismatch	F15	Resolve; BM02	M4	TBD
SR06	Playable publish incomplete	F16	Resolve; BM02	M4	TBD
SR07	Oversight/metrics incomplete	F17	Resolve; BM02	M4	TBD
SR08	Acceptance results absent	N01	Resolve; BM02	M2	TBD
SR09	Tag and CI unverified	Review	Resolve; BM02	M4	TBD

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F01	Accounts	Player	Verify then load own profile; reject invalid login.	F01 01–03	Partial
F02	Tutorial	Player	Complete/replay steps and retain status.	F02 01–03	Source
F03	Skill drills	Player	Validate trials; reward first eligible completion only.	F03 01–03	Source
F04	XP and unlocks	Player	Apply XP and unlocks once per command.	F04 01–03	Source
F05	AI coaching	Player	Validate adaptive challenge and labelled classifier quality.	F05 01–03	Partial
F06	Shots and rules	Player	Enforce serve, court, kitchen and scoring rules.	F06 01–03	Source
F07	Queue and ready	Player	Require all humans; decline safely cancels.	F07 01–03	Source
F08	Private rooms	Player	Validate coded room, roster and team readiness.	F08 01–03	Source
F09	Online play	Player	Synchronize permitted commands and disconnect policy.	F09 01–03	Partial
F10	Results and Elo	Player	Finalize agreed score and rating once.	F10 01–03	Source
F11	Stats and replay	Player	Read own history, errors and authorized replay.	F11 01–03	Source
F12	Friends and rank	Player	Authorize relationships and stable Elo ordering.	F12 01–03	Source
F13	Objectives	Player	Claim each eligible period reward once.	F13 01–03	Source
F14	Offline sync	Player	Practice offline then reconcile once online.	F14 01–03	Gap
F15	Admin access	Admin	Reject non-admin; persist valid access changes.	F15 01–03	Partial
F16	Content publish	Admin	Published valid config appears in game catalog.	F16 01–03	Partial
F17	Moderation	Admin	Audit report and appeal decisions.	F17 01–03	Partial
F18	Analytics	Admin	Use dated metric definitions and denominators.	F18 01–03	Partial

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
N01	Performance	Frame, input, feedback	60 FPS; p95 input ≤100 ms; feedback <1 s	Release device traces
N02	Network/load	RTT and concurrency	<100 ms VN; 2,000 players / 500 matches	Staging load + impairment
N03	Availability	Monthly uptime	≥99.5%; recovery targets to agree	Probe + restore/rollback
N04	Platform/UX	Main-flow matrix	Win10+, Android9+, iOS14+; no clipping	Physical device matrix
N05	Security	RBAC/data trust	JWT; no unauthorized award; AES-256/dedicated proof	Access, race, config tests
N06	Offline and learning	Sync and ML quality	Offline once-only sync; ML threshold to agree	Disconnect + labelled evaluation

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
M1 TBD	Identity training	Account and tutorials; Training and player progression	WF01/WF05	Proposed only
M2 TBD	Local gameplay	Rules and match presentation; Bot coaching and performance	WF02	Proposed only
M3 TBD	Online play	Sessions and rooms; Results and social integration	WF03/WF04	Proposed only
M4 TBD	Backend administration	Admin and catalog services; Deployment security and observability	WF06	Proposed only

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF01	Register learn and complete a first drill	Player, SMTP delivery service	Submit and verify account	A verified player completes a first drill and sees the persisted result.	F01,F02,F03	Account registered → verified → tutorial completed → training attempt completed	M1 TBD
WF02	Practice against an adaptive bot	Player	Choose bot difficulty and court	The player receives a legal match result and coaching evidence.	F05,F06,F10,F11	Match Intro → PreServe → Rally → MatchOver → result persisted	M2 TBD
WF03	Find accept and finish an online match	Online player A, Online player B, Photon service	Join eligible skill queue	A canonical match result updates each participant once.	F06,F07,F09,F10	Queue searching → offer → all accepted → playing → reported → finished	M3 TBD
WF04	Create a private team match	Room host, Invited player	Find a friend or share room code	A coded room starts with the declared 1v1 or 2v2 roster.	F08,F09,F10,F12	Room created → joined → teams ready → playing → finished	M3 TBD
WF05	Review progress and claim rewards	Player	Refresh server progress	The player sees next practice, history and exactly-once rewards.	F03,F04,F11,F13,F14	Attempt completed → progress refreshed → reward eligible → claimed	M1 TBD
WF06	Administer users content and operations	Administrator, Reporting player	Authenticate as administrator	Authorized actions change service data and support operational decisions.	F15,F16,F17,F18	User active → sanctioned/restored; lesson draft → published; report pending → resolved	M4 TBD

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
WF01-Main	WF01	Main	Seeded staging; named presenter pending	Submit and verify account; Open tutorial and perform controls; Choose an unlocked drill; Start server attempt and submit measurements; Complete and refresh stars and first reward	A verified player completes a first drill and sees the persisted result.	M1
WF01-Exception	WF01	Exception	Seeded staging; named presenter pending	Expired OTP or invalid trial data is rejected; account/attempt can be retried without duplicate reward.	Reject/preserve/retry per policy	M1
WF02-Main	WF02	Main	Seeded staging; named presenter pending	Choose bot difficulty and court; Run entry cinematic; Play serve and rally with legal inputs; Resolve points and match end; Wait for end notice then show player result and statistics	The player receives a legal match result and coaching evidence.	M2
WF02-Exception	WF02	Exception	Seeded staging; named presenter pending	Illegal kitchen/serve action is penalized; an aborted/unverified result cannot award a duplicate reward.	Reject/preserve/retry per policy	M2
WF03-Main	WF03	Main	Seeded staging; named presenter pending	Join eligible skill queue; Poll canonical offer and accept; Wait for all required human acceptances; Start Photon Shared session and play; Report consistent results and read finalized rating	A canonical match result updates each participant once.	M3
WF03-Exception	WF03	Exception	Seeded staging; named presenter pending	A decline/timeout cancels the offer; a conflicting report blocks normal finalization and applies server policy.	Reject/preserve/retry per policy	M3

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
WF04-Main	WF04	Main	Seeded staging; named presenter pending	Find a friend or share room code; Create selected room mode; Join and assign valid teams; Confirm ready roster and start; Finish and inspect authorized statistics	A coded room starts with the declared 1v1 or 2v2 roster.	M3
WF04-Exception	WF04	Exception	Seeded staging; named presenter pending	Wrong/full room or unauthorized team/start command is rejected; the room stays consistent.	Reject/preserve/retry per policy	M3
WF05-Main	WF05	Main	Seeded staging; named presenter pending	Refresh server progress; Inspect stars and error aggregates; Open suggested drill or replay; Claim eligible mission/achievement reward; Refresh wallet and continuation target	The player sees next practice, history and exactly-once rewards.	M1
WF05-Exception	WF05	Exception	Seeded staging; named presenter pending	A repeated claim does not credit again; offline practice/sync is a required gap and must be demonstrated after implementation.	Reject/preserve/retry per policy	M1
WF06-Main	WF06	Main	Seeded staging; named presenter pending	Authenticate as administrator; Inspect a user or moderation report; Apply valid access/status decision; Create and publish playable lesson configuration; Inspect period-defined operational analytics	Authorized actions change service data and support operational decisions.	M4
WF06-Exception	WF06	Exception	Seeded staging; named presenter pending	A non-admin request is denied; invalid publication/status data leaves existing records unchanged.	Reject/preserve/retry per policy	M4

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	18	3	54
Main workflows	6	1	6
State/guard transitions	118	1	118
NFR groups	6	1	6
Proposed total	184	Not signed	184

At least 92/184 must actually run in CI after agreement. Current actual results: NOT RUN. Main demos 6/6; exception ≥5/6; overall test pass ≥166/184 plus 100% mandatory main paths. No open Critical/High at Defense 1.1.